INCH-POUND
MIL-PRF-62187A
30 December 1996
SUPERSEDING
MIL-R-62187
10 February 1988
(See 6.5)

PERFORMANCE SPECIFICATION

REGULATOR, 650 AMP, SOLID STATE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a solid state voltage regulator for use primarily on military combat vehicles.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2920

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SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-G-62186	- Generator, Alternating Current (650 Ampere Rectified), 28 Volts Direct Current (Oil Cooled).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

11655469	- Generator AC/DC (Oil Cooled).
12354334	- Regulator, Solid State.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC Z1.4	- Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).
ANSI/SAE AS478	- Identification and Marking Methods (DoD Adopted).

(Application for copies should be addressed to American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A123	- Standard Specification for Zinc (Hot Dip Galvanized) Coatings on Iron and Steel Products (DoD Adopted).

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

GENERAL MOTORS CORPORATION

GM 9540P	- Accelerated Corrosion Test.

(Copies of GM publications may be obtained from General Motors Corporation, c/o Global Engineering, 15 Inverness Way, Englewood, CO 80112.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.4.

3.2 Materials. Materials shall be as specified herein, in applicable specifications, and on applicable drawings. Materials not specifically designated shall be of good commercial quality. Materials used shall be suitable for use in the regulator operating over an ambient temperature range of -65 to 225 degrees Fahrenheit (°F) (-54° to +107 degrees Celsius (°C)) without any change in physical or dimensional properties that would result in operation of the regulator falling outside of the limits specified (see 4.6.1).

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3.2.1 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals which could cause galvanic action, shall be avoided. Separation of dissimilar metals shall be accomplished by providing appropriate insulation between mating surfaces.

3.2.2 External surfaces. All external surfaces shall be corrosion resistant (see 4.6.1). Corrosion protection shall be equal to or exceed that provided by hot dip galvanized 1010 steel with coating thickness in accordance with ASTM A123 with zinc phosphate pretreatment (see 4.8.3).

3.2.3 Fungicidal materials. All non-metallic materials shall be inherently fungicidal or treated to resist fungus growth except when used for components of potted compounds or otherwise moisture sealed assemblies (see 4.6.1).

3.2.4 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design and construction. Regulators acquired in accordance with this specification shall conform to Drawing 12354334 and specifications and standards to the extent specified herein (see 4.6.1 and 4.6.2).

3.3.1 Interchangeability. All regulators shall be interchangeable with respect to installation and performance. The voltage regulator after calibration at time of manufacture or overhaul shall not require any adjustment at time of installation. To provide for interchangeability, the assembly fabricated in accordance with this specification shall have external dimensions conforming to the envelope outline of Drawing 12354334. Electrical connectors, mounting dimensions, and locations shall conform to the detail requirements of Drawing 12354334. Electrical contact designation shall conform to requirements of Drawing 12354334 (see 4.6.1 and 4.6.2).

3.3.2 Electrical interface. Electrical connector pin designations to the utilizing vehicle shall be provided as follows (see 4.6.2):

Connector pin function	Connector pin designations
Regulation Sensing Input	A
Current Limit Signal Input	B
Ground	C
Generator Alternator AC Signal Input	D

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Connector pin function	Connector pin designations
Generator Alternator Field Control Output	E
Accessory Relay Signal Output	F
Overvoltage Test Circuit Input	G

3.3.3 Standard hardware. Military standard parts shall be used wherever possible (see 4.6.1).

3.3.4 Generator-alternator compatibility. The regulator shall function as specified with generator-alternators conforming to MIL-G-62186 and Drawing 11655469 (see 4.6.1 and 4.7).

3.3.5 Installation compatibility. The regulator shall fit and perform as specified herein when a trial installation with the generator-alternator (see 3.3.4) is specified on the M60A3 tank (see 4.6.1 and 6.2).

3.3.6 Weight. The regulator shall not exceed a weight of 6 pounds (2.7 kilograms (kg)) (see 4.6.2).

3.4 Performance.

3.4.1 Voltage regulation. The regulator shall establish and maintain output voltage in accordance with figures 1 and 4 throughout the generator-alternator speed range of 2400 to 8000 revolutions per minute (rpm) (see 4.7.1).

3.4.2 Ripple voltage. Ripple voltage shall not exceed 4 volts (V) peak to peak with output voltage at 28 V direct current (dc) throughout the generator- alternator speed range of 2400 to 8000 rpm (see 4.7.2).

3.4.3 Overvoltage. The overvoltage portion of the regulator shall deactivate the field circuit when the system voltage (Es) reaches 33.0 ± 1.0 V. Deactivation and return to battery voltage shall be accomplished in 0.07 seconds. Reset shall be manual and shall be located on the regulator (see 4.7.3).

3.4.4 False trips. False tripping of the regulator shall not be caused by buildup, load switching, and input speed acceleration or deceleration (see 4.7.4 and 4.7.7.1).

3.4.5 Current limit. The current limiting shall be incorporated into the regulator assembly which will limit the generator-alternator output current controlled by the regulator in accordance with figure 2. Stable current limiting shall be established within 0.05 seconds. System voltage undershoot shall not be below 18 V (see 4.7.5).

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3.4.6 Accessory relay control. The accessory relay shall close when the generator-alternator speed is between 900 and 1100 rpm. This relay shall open before the generator-alternator speed reaches 800 rpm during shutdown (see 4.7.6).

3.4.7 Fault protection. Circuitry shall be included in the regulator to deactivate the field circuit in the event of wiring malfunctions. The circuitry shall not operate if a malfunction in any of the trip types of table I does not create a condition which would damage the regulator or associated electrical power sources. Maximum voltage (V), amperes (A) and trip time limits shall be as specified in table I for the three trip types (see 4.7.7).

TABLE I. Trip types.

Trip type	Generator-alternator rpm	E_A V	E_S V	I_F A	Time seconds
Regulator sensing	2000	35.0	35.0	3	0.06
Input lead	8000	35.0	35.0	4	1.00
Alternator positive	2000	40.0	30.0	4	1.50
Output lead	8000	115.0	30.35	2	1.70
Alternator negative	2000	35.0	30.5	5	2.0
Output Lead	8000	55.0	30.5	1	0.4

3.4.8 Dielectric strength. The regulator circuit insulation to ground shall not permit a leakage current greater than 5 milliamperes (mA) with 500 V alternating current (ac) applied between all receptacle pins at the same time and the case with the case ground lead disconnected (see 4.7.8).

3.4.9 Short circuit protection. Regulator circuitry shall be provided to prevent damage to internal components in the event of a short circuit from pin "E" or "F" to ground during normal regulator operation (see 4.7.9).

3.4.10 Endurance. The regulator shall meet the following requirements before and after 3000 hours (125 cycles) of operation conforming to the time schedule and conditions set forth in table II. The regulator shall subsequently meet the requirements of 3.4.1 through 3.4.3, 3.4.5 through 3.4.9, 3.4.12, and 3.5.5 (see 4.7.10 and 4.7.10.1).

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TABLE II. Regulator endurance time schedule and conditions.

1/ Time, hours	2/ RPM	3/ System voltage, volts	Alternator resistive load			Load cycles, minutes	Cyclic load, seconds	
			Constant current, A	Cyclic current, A	Total current, A		On	Off
1.5	2400	26.5 ±0.75	320	330	650	90	3	57
7.5	5120	26.5 ±0.75	320	330	650	450	3	57
14.5	3520	26.5 ±0.75	320	330	650	870	3	57
0.5	8000	26.5 ±0.75	650	0	650	30	--	--

1/ 24 hours per cycle.
2/ Alternator.
3/ System voltage in table II shall be applied at +225°F (see 3.4.7). See figure 4 for voltage values at lower temperatures.

3.4.11 Electromagnetic interference. The regulator shall not exceed the broadband, conducted and radiated emission limits of figure 6 and 7 (see 4.7.11 and 6.1).

3.4.12 Case grounding. The regulator case shall be electrically connected to the ground circuitry (see 4.7.12).

3.4.13 Reverse polarity protection. The regulator circuit shall not be damaged in the event reverse polarity is applied to the input power connector (see 4.7.13).

3.5 Environmental conditions. The regulator shall meet the requirements of 3.4 after being subjected to the conditions specified herein.

3.5.1 Shock resistance. When mounted as in intended use, the regulator shall withstand 40 gravities (g) peak sawtooth shock pulses for an 18 ±3 millisecond (ms) duration in each of the major mutually perpendicular axes (see 4.8.1 and 4.8.1.1).

3.5.2 Vibration resistance. The assembly, when mounted as intended use, shall withstand a maximum input vibration level of 4 g at a sweep frequency range of 5 to 500 Hertz (Hz) for a period of 100 minutes in each of three mutually perpendicular axes (see 4.8.2 and 4.8.2.1).

3.5.3 Corrosion resistance. The regulator shall meet performance requirements of 3.4 after exposure to 192 hours of salt fog (see 4.8.3).

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3.5.5 Waterproofness. The assembly shall withstand 10 pounds per square inch gage (psig) (68.95 kPa (kilopascals) internal pressure and 10 psig internal vacuum without leakage and while operating in salt water (see 4.8.5).

3.5.6 Fungus resistance. The regulator shall meet the requirements of 3.4 after 28 days exposure to fungal incubation (see 4.8.6).

3.5.7 Extreme temperature resistance. The regulator shall meet the performance requirements of 3.4 during and after operation at extreme ambient temperatures shown below. The range of regulated voltage over the total operational temperature range shall conform to figure 4 (see 4.8.7).

a. High temperature: +225°F.
b. Low temperature: -65°F.

3.6 Identification and marking. Identification and marking shall be in conformance with ANSI/SAE AS478 and MIL-P-15024 (see 4.6.2).

3.6.1 Nameplate. A nameplate shall be attached to the regulator housing in the manner specified on the applicable standard or drawing. The legend contained on the nameplate shall include the following as a minimum (see 4.6.2).

a. Regulator, solid state
b. 28 volts dc
c. Army part number
d. National stock number
e. Manufacturer's identification (CAGE)
f. Manufacturer's serial number
g. Contract number
h. Date of manufacture
i. US

3.7 Workmanship. The regulator shall be free of rust, burrs and scratches. Solder connection processing is optional but must meet all electrical, mechanical and environment requirements herein. All solder cleaning agents and flux shall be chemically neutralized and removed after soldering. Manufacturing methods shall not cause degradation of the inherent reliability and durability of the regulator (see 4.6.2).

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4. VERIFICATION

4.1. Inspection equipment. Unless otherwise specified (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment shall be capable of repetitive measurements to an accuracy of 10 percent (%) of the measurement tolerance.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Conformance inspection (see 4.5).

4.3 Inspection conditions.

4.3.1 Atmospheric limits. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 77 ±15°F (25±7°C)
b. Barometric pressure: 28.5 (+2, -3) inches of mercury (96.5 (+6.8, -10 kPa)).
c. Relative humidity: 50 ±30%

4.3.2 Laboratory test conditions. Tests shall be conducted under general laboratory conditions acceptable to the Government with the following procedures:

a. Oscillographics shall have a frequency response from 0 to 2000 Hertz which is flat within ±5%.
b. Prior to each performance test, the regulator shall be operated for 15 minutes at 3000 rpm at 650 A alternator current for temperature stabilization.
c. Electrical switches SW8, SW9 and SW13 (see figure 3) shall be closed at all times unless otherwise specified, and SW3 shall always be open unless otherwise specified. SW1 shall be the first closed when energizing the regulator followed by SW2. SW4 shall be closed whenever resistive load is required during these tests. When de-energizing the system, SW1 shall always be the last switch opened unless otherwise specified. SW1 shall be opened and closed only with the rotary power source at rest.

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select four regulator assemblies produced under the production contract for first article inspection. First article testing (see 3.1) shall be conducted by the contractor under Government surveillance, and shall consist of defects specified in 4.6.2 and tests specified in table III.

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Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply regulator assemblies that are fully representative of those inspected as a first article sample.

4.4.1 Preproduction inspection (PPT). When specified (see 6.2), the preproduction sample shall consist of two regulators. Preproduction inspection shall consist of inspection as specified in table III.

4.4.2 Initial production inspection (IPT). Unless otherwise specified (see 6.2), the Government shall select two regulators from the first ten produced under the production contract for initial production inspection. Initial production units shall be inspected as specified in table III.

4.4.3 First article inspection failure. Test item deficiencies found during, or as a result of the first article test shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article test, shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article testing is completed to the satisfaction of the Government.

4.5 Conformance inspections.

4.5.1 Sampling.

4.5.1.1 Lot formation. Unless otherwise specified (see 6.2), an inspection lot shall consist of all the regulator assemblies of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for conformance examinations shall be selected from an inspection lot in accordance with ANSI/ASQC Z1.4.

TABLE III. Classification of inspections.

Title	Requirement	Inspection	First article		Conformance		Control tests
			PPI	IPT	Examination	100% Tests	
Materials and construction	3.2 thru 3.2.4, 3.3.4 and 3.3.5	4.6.1	X	X			1/300
Defects	3.3 thru 3.3.3 3.3.6, 3.6 and 3.7	4.6.2	X	X	X		

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TABLE III. Classification of inspections - Continued.

Title	Requirement	Inspection	First article		Conformance		Con-trol tests
			PPI	IPT	Exami-nation	100% Tests	
Voltage regulation	3.4.1	4.7.1	X	X			
Voltage regulation	3.4.1.1	4.7.1.1	X	X		X	
Ripple voltage	3.4.2	4.7.2	X	X			
Ripple voltage	3.4.2	4.7.2.1	X	X		X	
Current limit	3.4.5	4.7.5	X	X			1/300
Current limit	3.4.5	4.7.5.1	X	X		X	
Accessory relay	3.4.6	4.7.6	X	X			
Fault protection	3.4.7	4.7.7	X	X			
Fault protection	3.4.7	4.7.7.1	X	X		X	
Short circuit protection	3.4.9	4.7.9	X	X			
Overvoltage	3.4.3	4.7.3	X	X			
Overvoltage	3.4.3	4.7.3.1	X	X		X	
Dielectric strength	3.4.8	4.7.8	X	X			
Accessory relay	3.4.6	4.7.6	X	X		X	
Short circuit protection	3.4.9	4.7.9	X	X		X	
Dielectric strength	3.4.8	4.7.8	X	X		X	
Case grounding	3.4.12	4.7.12	X	X		X	
Reverse polarity protection	3.4.13	4.7.13	X	X		X	
Waterproofness	3.5.5	4.8.5	X	X			1/50
Waterproofness	3.5.5	4.8.5.1	X	X		X	
Shock resistance	3.5.1	4.8.1	X	X			
Control shock resistance		4.8.1.1					1/50
Vibration resistance	3.5.2	4.8.2	X	X			
Control vibration resistance		4.8.2.1					1/50
Corrosion resistance	3.5.3	4.8.3	X	X			
Humidity	3.5.4	4.8.4	X	X			
Fungus resistance	3.5.6	4.8.6	X	X			
High temperature	3.5.7	4.8.7	X	X			1/100
Low temperature	3.5.7	4.8.7	X	X			1/100
Endurance	3.4.10	4.7.10	X	X			
Endurance	3.4.10	4.7.10.1					1/300
Electromagnetic interference	3.4.11	4.7.11	X	X			

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4.5.2 Examinations. Regulators shall be examined for the defects specified in table IV.

TABLE IV. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major:		
101	Materials not certified (see 3.2).	Visual
102	Incorrect dimensions affecting mounting and interchangeability (see 3.3 and 3.3.1).	SIE 1/
103	Contacts other than specified (see 3.3.2).	Visual
104	Non-military standard parts where military standard parts are applicable (see 3.3.3).	Visual and SIE
105	Non-conforming weight (see 3.3.6).	SIE
106	Faulty workmanship affecting performance (see 3.7).	Functional
Minor:		
201	Incorrect dimensions not affecting interchangeability (see 3.3.1).	SIE
202	Improper marking and nameplates (see 3.6).	Visual
203	Faulty workmanship affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Test (100%). Each regulator shall be subjected to the conformance tests specified in table III.

4.5.4 Control inspection. Unless otherwise specified, inspections shall be conducted by the supplier for maintaining control of those requirements whose length, expense, or destructiveness prohibit their inclusion under lot-by-lot inspection. Samples shall be selected and subjected to the inspections with the frequencies designated in table III. These inspections need not be performed on the first control inspection quantity when initial production inspection has been satisfactorily accomplished.

4.6 Methods of inspection.

4.6.1 Materials and construction. Conformance to 3.2 and 3.3 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

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4.6.2 Defects. Conformance to 3.2 through 3.7 shall be determined by examination for the defects listed in table IV. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.7 Performance. Regulator shall meet the following requirements using a generator-alternator conforming to MIL-G-62186 (see 3.3.4).

4.7.1 Voltage regulation. To determine conformance to 3.4.1, the regulator shall be installed in a circuit as shown in figure 3 and the regulator shall be operated. With SW1 and SW2 closed, restrictive loads of zero (battery only), 100A and 650A shall be applied and the steady-state regulated voltage (E_s) shall be recorded. These recordings shall be made at generator-alternator speeds of 2400, 4000, 5000 and 8000 rpm and shall be in accordance with figure 4. With a resistive load of 650A suddenly applied and removed (application and release times shall not exceed 10 ms), oscilligraphs of the transient regulated voltage (E_s) shall be taken at generator-alternator speeds of 2400, 4000, 6000 and 8000 rpm and shall be in accordance with figure 1.

4.7.1.1 Abbreviated voltage regulation. With the stabilized regulator installed in a circuit as in 4.7.1, the output voltage shall be observed during the following procedures. The resistive load shall be applied and removed suddenly as indicated.

a. Operate generator-alternator at 2400 rpm with resistance load at zero (battery only).
b. With generator-alternator at 2400 rpm, add resistance load to establish a total current flow of 100 A.
c. With generator-alternator speed at 2400 rpm, increase resistive load to establish a current flow of 650 A.
d. Increase generator-alternator speed to 4000 rpm, and remove the resistance load.
e. With generator-alternator speed at 4000 rpm add resistance load to establish a total current flow of 100 A.
f. With generator-alternator at 4000 rpm, increase resistance load to establish a current flow of 650 A.
g. Increase generator-alternator speed to 6000 rpm and remove the resistance load.
h. With generator-alternator at 6000 rpm, add resistance load to establish a total current flow of 100 A.
i. With generator-alternator speed at 6000 rpm, increase resistive load to establish a current flow of 650 A.
j. Increase generator-alternator speed to 8000 rpm and remove the resistance load.
k. With generator-alternator at 8000 rpm, add resistance load to establish a total current flow of 100 A.

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1. With generator-alternator speed at 8000 rpm, increase resistive load to establish a current flow to 650 A.

4.7.2 Ripple voltage. To determine conformance to 3.4.2, the regulator shall be connected in a circuit as shown in figure 3 and operated at the rate and load to produce a current flow as indicated below. Ripple voltage shall be measured at point E_s and oscillograph readings recorded for these conditions for peak to peak and +peak to -peak.

Rpm	Current, A (I_A)
3000	50
3000	650
8000	50
8000	650

4.7.2.1 Abbreviated ripple voltage. The regulator shall be tested as in 4.7.2 except that the test point shall be 8000 rpm at a load producing a current flow of 650 amperes (I_A).

4.7.3 Overvoltage. To determine conformance to 3.4.3, the regulator shall be connected in a circuit as shown in figure 3 and shall be operated as follows. With SW1 and SW2 closed adjust alternator speed to 5000 rpm. Observe system voltage (E_s) on an oscillograph as SW3 is closed. Observe E_s where trip occurs. Open SW3 and reset overvoltage switch on regulator. Read the trip voltage and time required for a return to battery voltage from the oscillograph.

4.7.3.1 Abbreviated overvoltage. With the regulator connected as shown in figure 5, the power supply voltage shall be increased until overvoltage circuit breaker trips. The dc voltmeter reading shall be recorded only at the overvoltage switch trip point. The dc power shall be disconnected and the regulator overvoltage switch shall be reset.

4.7.4 False trips. Conformance to 3.4.4 shall be determined to be successful after completion of all Performance and Environmental tests of sections 4.7 and 4.8 without erroneous tripping of the regulator (see 4.7.7.1).

4.7.5 Current limit. To determine conformance to 3.4.5, the regulator shall be installed in a circuit such as shown in figure 3 and the following test shall be performed:

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a. Current limit, transient. Open switch SW13 and operate the regulator for 15 minutes with a 500 ampere alternator current flow at 5000 rpm. Disconnect all load from the alternator except batteries. Connect a resistive load of 0.028 ohms. Alternator current (I_A) and system voltage (E_S) shall be observed on an oscillograph tape and read to determine conformance with the recovery and undershoot requirements of 3.4.5.

b. Current limit, regulation. Apply a resistive load to produce a current flow of 50 A from the alternator at 2400 rpm. Increase the loading in 50A increments until the system voltage is reduced to less than 24 V. Measure the system voltage and current to determine conformance to figure 2. Repeat at 5000 rpm alternator speed.

4.7.5.1 Abbreviated current limit. With the regulator installed as in 4.7.5, and operationally stabilized and with the alternator at 2400 rpm, a resistive load shall be applied to produce a current flow of 400 A from the alternator and output at E_s shall be recorded. The load shall be increased until E_s is 26 ± 0.1 V and shall be measured to verify that it conforms to the limits of figure 2. This test shall be repeated at an alternator speed of 5000 rpm.

4.7.6 Accessory relay control. To determine conformance to 3.4.6, the regulator shall be tested while it is installed in a circuit as is shown in figure 3. With SW1 and SW2 closed, the alternator speed shall be slowly increased from zero rpm until the accessory relay closes (light at L1 comes on). Now the alternator speed shall be slowly decreased until the accessory relay opens (light L1 goes out). Alternator speeds at which relay closes and opens shall be recorded.

4.7.7 Fault protection. The regulator shall be installed in a circuit as shown in figure 3. The following tests shall be performed to determine conformance to 3.4.7. Measure E_s, E_A and I_f on an oscillograph for each of the faults described below:

a. Regulator sensing input lead.
   1. Operate the regulator and alternator at 2000 rpm with no external alternator load with SW10 closed. Set variable resistance R10 at zero ohms and open SW10. Slowly increase R10 until the protective circuit trips. Measure E_A, E_s and I_f just before the trip occurs. Leave R10 at the final setting. Close SW10 and reset the regulator overvoltage switch.
   2. Operating at 2000 rpm, open SW10 while observing E_A, E_s and I_f on an oscillograph. After the circuit trips, close SW10 and reset regulator overvoltage switch.
   3. Repeat a1 above at 8000 rpm.
   4. Repeat a2 above at 8000 rpm.

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b. Alternator positive output cable.
   1. Operate the regulator and alternator at 2000 rpm, with no external alternator load with SW11 closed. Set R11 at zero ohms and open SW11. Slowly increase R11 until the protective circuit trips. Measure E_A, E_S and I_f just before the trip occurs. Leave R11 at the final setting. Close SW11 and reset overvoltage switch.
   2. Operating at 2000 rpm, open SW11 while observing E_A, E_S and I_f on an oscillograph. After the circuit trips, close SW11 and reset the overvoltage switch.
   3. Repeat b1 above at 8000 rpm.
   4. Repeat b2 above at 8000 rpm.

c. Alternator negative output cable.
   1. Operate the regulator and alternator at 2000 rpm with no external alternator load with SW12 closed. Set R12 at zero ohms and open SW12. Slowly increase R12 until the protective circuit trips. Measure E_A, E_S and I_f just before the trip occurs. Leave R12 at the final setting. Close SW12 and reset the overvoltage switch.
   2. Operating at 2000 rpm, open SW12 while observing E_A, E_S and I_f. , on an oscillograph. After the circuit trips, close SW12 and reset the overvoltage switch.
   3. Repeat c1 above at 8000 rpm.
   4. Repeat c2 above at 8000 rpm.

4.7.7.1 Abbreviated fault protection circuitry. To determine conformance to 3.4.4, operate the regulator with an external alternator load. The overvoltage switch shall trip after each step when steps a and b are performed in sequence (see 4.7.4).

a. Open SW8.
b. Close SW8, reset overvoltage switch and open SW9.

Then, close SW9 and reset the overvoltage switch. The regulator shall operate without tripping.

4.7.8 Dielectric strength. To determine conformance to 3.4.8, the following test shall be performed: Remove the cover and disconnect the internal circuit ground lead from the case. Connect all the pins of the regulator receptacle together. Connect a megohm meter between the shorted pins and the bare metal case. Starting with the tester voltage at 0 V, slowly increase the voltage to 500 V ac for 1 minute and then decrease the tester voltage to zero before removing the test leads and verify that the resistance is greater than 100 k ohms. Reconnect the circuit ground lead to the case after test and replace the cover.

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4.7.9 Short circuit protection. To determine conformance to 3.4.9, the regulator shall be connected to the test configuration as shown in figure 8 and the regulator shall be operated in accordance with table V.

TABLE V. Short circuit protection.

SW1	SW2	Circuit Breaker 1/	L1
Open	Open	On	On
Close	Open	Off	Off
Open	Open	Off	Off
Open	Open	On	On
Open	Close	Off	Off
Open	Open	Off	Off
Open	Open	On	On

1/ Breaker will automatically trip off, but must be reset by switching on by hand.

A test shall be considered successful when table V is followed in the order shown with the resulting circuit breaker and lamp (L1) response. The regulator shall successfully pass test 4.7.1 before and after this short circuit test is performed to insure no circuit has been damaged.

4.7.10 Endurance. To determine conformance to 3.4.10, the regulator shall be subjected to 3000 hours of operation per table II at an ambient temperature of 225 ±5°F. The following parameters shall be observed at the beginning and end of each speed change. After endurance testing, the regulator shall be tested in accordance with 4.7.1. Record the following data:

a. I_A alternator current.
b. Alternator rpm.
c. I_f field current.
d. E_f field voltage.

During the endurance test, the regulator shall require no servicing, adjustment, or replacement of parts.

4.7.10.1 Control endurance. To determine conformance to 3.4.10 for control inspections, the selected regulator shall be installed in a circuit as shown in figure 3, and operated for 4 sequential cycles per table II for a total of 96 hours. During this test the regulator shall require no servicing adjustments or replacements of parts. The regulator shall then be tested per 4.7.1.1.

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4.7.11 Electromagnetic interference (EMI). To determine conformance to 3.4.11, the regulator shall be interconnected with an alternator, conforming to Drawing 11655469 and shall be subjected to EMI tests for radiated and conducted broadband emissions in accordance with MIL-STD-462, method CE102 and RE102 using the limits shown in figures 6 and 7, respectively (see 6.1).

4.7.12 Case grounding. To determine conformance to 3.4.12, continuity between the regulator case and pin C of the connector shall be measured and shall be less than 1 ohm while the regulator is disconnected from the system.

4.7.13 Reverse polarity protection. To determine conformance to 3.4.13, 24 V dc shall be applied to pin C and ground shall be connected to pin A for 1 minute with the regulator circuit breaker switched on. The regulator shall meet performance test 4.7.1 before and after applying reverse polarity voltage to insure no circuits have been damaged due to testing.

4.8 Environmental conditions.

4.8.1 Shock resistance. To determine conformance to 3.5.1, the assembly shall be mounted as in use and shall be subjected to a total of 18 shock pulses in accordance with of MIL-STD-810, method 516.4, for ground equipment.

4.8.1.1 Control shock resistance. The regulator will be tested as in 4.8.1 except that the shock pulse amplitude shall be 30g.

4.8.2 Vibration resistance. To determine conformance to 3.5.2, the regulator shall be mounted by normal mounting provisions to a vibration exciter capable of producing and of transmitting the vibrations specified herein. The regulator shall be connected as in use and also connected to the instrumentation necessary for this test. Vibration shall be for 100 minutes in each direction along each of three perpendicular axes. Amplitude shall be 4 gravity units. Frequency shall be 5 to 500 to 5 Hz in a 15 minute time sweep. Test level shall be in accordance with the curve of figure 9. Resonant frequencies of the regulator shall be determined and a sinusoidal resonance search shall be conducted for the four most severe resonant frequencies of the unit. The regulator shall be subjected to 16.34 minutes at each severe resonant frequency within each 100 minutes sweep test. The frequency of applied vibration shall be swept over the specified range in accordance with figure 10. The sweep time is that of an ascending plus a descending time shown on figure 10. The vibration resistance test shall be performed at an ambient temperature of -65°F and 225°F.

4.8.2.1 Control vibration resistance. For control testing, the regulator will be tested as in 4.8.2 except that dwell time at each resonance shall be 2 minutes and vibration shall be performed at room ambient temperature.

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4.8.3 Corrosion resistance. To determine conformance with 3.2.2 and 3.5.3, the regulator shall be exposed to the accelerated corrosion test, GM 9540P, Method B except that the duration shall be for four test cycles (192 hours total). The regulator assembly shall be sealed so that internal components are not exposed.

4.8.4 Humidity resistance. To determine conformance to 3.5.4, the regulator shall be exposed to the following procedure:

a. The regulator shall be installed as in use in a chamber vented and baffled to assure an even flow of warm humid air as specified herein. No condensation shall be permitted to drip on the regulator. Relative humidity shall be measured by the dry bulb-wet bulb thermometer comparison method or approved equivalent. Air velocity in the internal test space shall not exceed 150 feet (45.6 meters) per minute. Steam or distilled, demineralized or deionized water having a pH value between 6 and 7.2 at 73 ±3°F (23 ±1°C) shall be used to obtain the specified humidity. No rust or corrosive contaminants shall be imposed on the test sample by the test facility. During this test, no repair or replacement shall be permitted and the regulator shall be operated only when necessary.
b. After having been dried for 24 hours at 129 ±3°F (54 ±1.5°C), the regulator shall be conditioned at 73 ±3°F and 50 ±10% relative humidity for 24 hours. The internal chamber temperature shall be raised to 86 ±3°F (30 ±2°C) and 94 ±4% relative humidity. The regulator shall be exposed to 5 continuous 48 hour cycles in accordance with figure 11. Accumulated moisture may be removed by turning the regulator upside down or shaking. Wiping shall not be permitted. The regulator shall have no visible damage, corrosion or swelling as a result of exposure to humidity. The regulator shall subsequently meet performance requirements by testing as specified in 4.7.1.

4.8.5 Waterproofness. To determine conformance to 3.5.5, the regulator shall be subjected to internal conditions of 10 psig pressure. Using a 1/8 inch adapter fitting, a vacuum pump shall be connected to the regulator. Temperature of the liquid and test sample shall be 73 ±18°F (23 ±10°C). The regulator shall be submerged into a salt water bath (see 4.8.3) with a minimum depth of 1 inch (2.54 cm) covering the unit. The following procedure shall be performed: A 10 psig (68.9 kPa) vacuum shall be applied and the regulator operated in accordance with 4.7.1. After the vacuum test, remove unit from water bath and thoroughly dry outside of test sample. Carefully remove the cover and inspect the regulator for water leakage. Evidence of moisture in or on the regulator interior shall not be acceptable. Carefully replace cover, remove the vacuum pump and connect to a dry nitrogen source. Submerge the regulator as before, apply 10 psig dry nitrogen pressure and test in accordance with 4.7.1. Escaping bubbles from the regulator interior during this test shall not be acceptable.

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4.8.5.1 Abbreviated waterproofness. For conformance testing, the regulator shall be subjected to the test of 4.8.5 except that operation and vacuum testing are not required. A pressurized test shall be conducted using air and after submergence the regulator shall conform to 4.7.1.1.

4.8.6 Fungus resistance. To determine conformance to 3.5.6, the regulator shall be subjected to fungal incubation for 28 days in accordance with method 508.4 of MIL-STD-810. Samples shall be incubated under cyclic temperature and humidity conditions and shall include 20 hours of relative humidity at 95 ±5% at an air temperature of 86 ±2°F (30 ±1°C) followed by 4 hours of 100% relative humidity at 77 ±2°F (25 ±1°C).

4.8.7 Extreme temperature resistance. To determine conformance to 3.5.7, test the regulator in a chamber suitable for the development and maintenance of the specified temperatures in accordance with MIL-STD 810, methods 501.3 and 502.3. The regulators shall be tested in accordance with 4.7.1 before, during and after each test specified below.

a. Low temperature.
   1. Storage: After stabilization in the chamber, the regulator shall be kept at -80°F (-62°C) for 12 hours.
   2. Operation: After storage at -80°F, the temperature shall be raised to -65°F. When stabilized at this temperature, the regulator shall be tested as specified in 4.7.1.1, 4.7.2.1, 4.7.3.1, 4.7.4, 4.7.5.1, 4.7.6, 4.7.7.1, 4.7.9, 4.7.10, 4.7.12, and 4.7.13. Voltage values shall conform to figure 4.

b. High temperature.
   1. With the regulator installed as ready to operate, the internal chamber temperature shall be raised to 160°F (71°C) within 1 hour time period. This temperature shall be maintained for an additional 4 hours. The internal chamber temperature shall then be lowered to 120°F (49°C) within a 1 hour time period. This cycle shall be repeated three times for a total of 36 hours. The internal chamber temperature shall be adjusted to 225°F and maintained until the regulator is temperature stabilized. The regulator shall be operated and tested in accordance with 4.7.1.1, 4.7.2.1, 4.7.3.1, 4.7.4, 4.7.5.1, 4.7.6, 4.7.7.1, 4.7.9, 4.7.10, 4.7.12, and 4.7.13. Voltage values shall conform to figure 4. The regulator shall be returned and stabilized to standard ambient conditions. No operational deficiencies or physical damage shall exist when the regulator is operated.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging Data retrieval is available from the managing Military Departments or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The regulator is intended to control and monitor the alternator to provide prescribed electrical power for engine driven military surface vehicles. The EMI requirements herein are applicable only to the use of this item in M60 Series Tanks (see 3.4.11).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. When first article is not required (see 3.1).
d. If trial installation on an M60A3 tank is required (see 3.3.5).
e. If responsibility for inspection equipment should be other than as specified (see 4.1).
f. If inspection conditions should be other than as specified (see 4.3.1).
g. If the number of first article samples should be other than as specified (see 4.4).
h. If preproduction inspection is required (see 4.4.1).
i. If initial production inspection is required (see 4.4.2).
j. If the inspection lot samples should be other than as specified (see 4.5.1.1).
k. Packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 3.2.4 and 6.3.2).

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6.3.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.). or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.4 Subject term (key word) listing.

Direct current regulator
Electrical power
M60 series tank
Steady state
Voltage

6.5 Supersession data. This military specification supersedes purchase description ATPD MIL-P-62187(AT), dated 11 December 1973, and MIL-R-62187, dated 10 February 1988.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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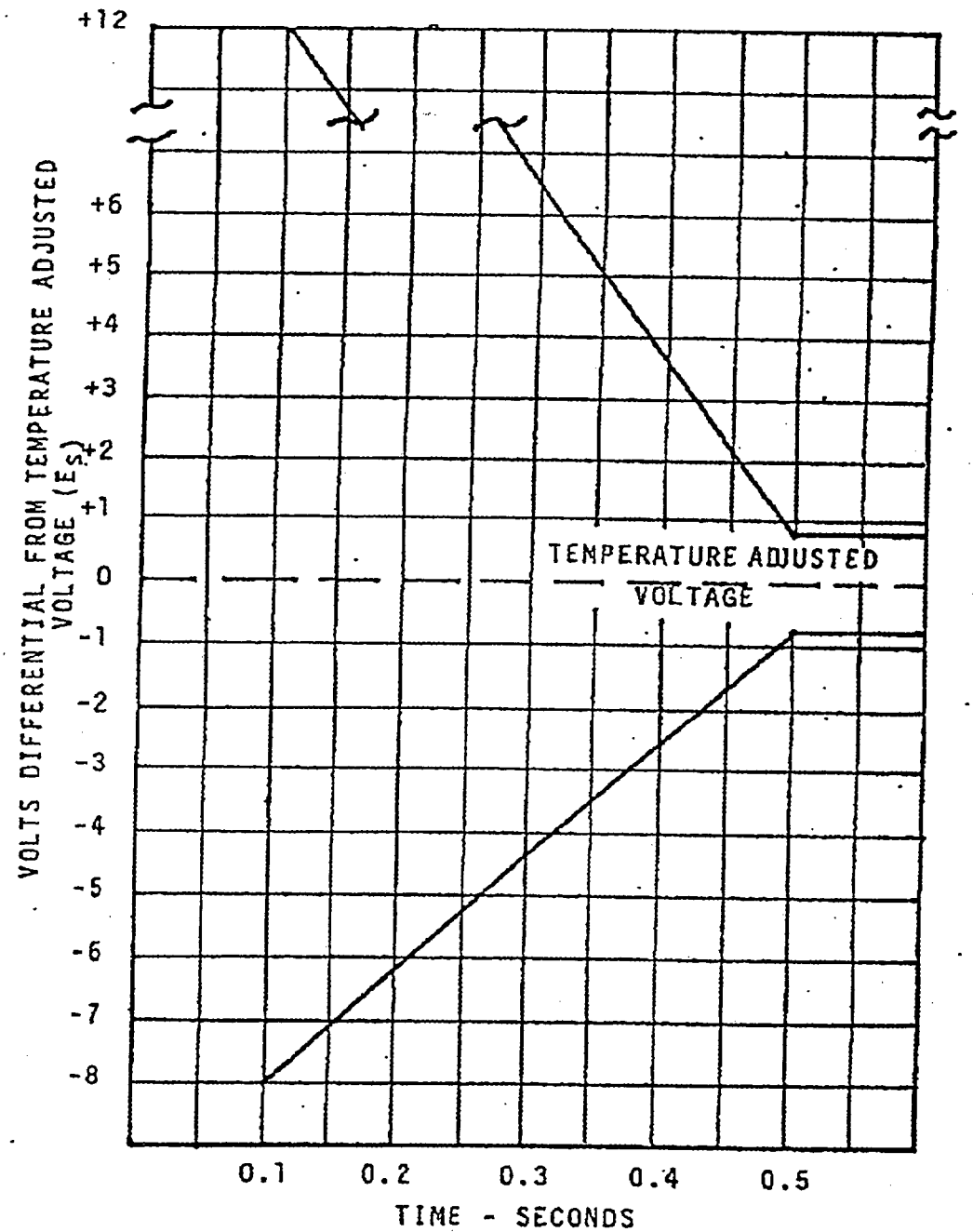


FIGURE 1. Voltage regulation.

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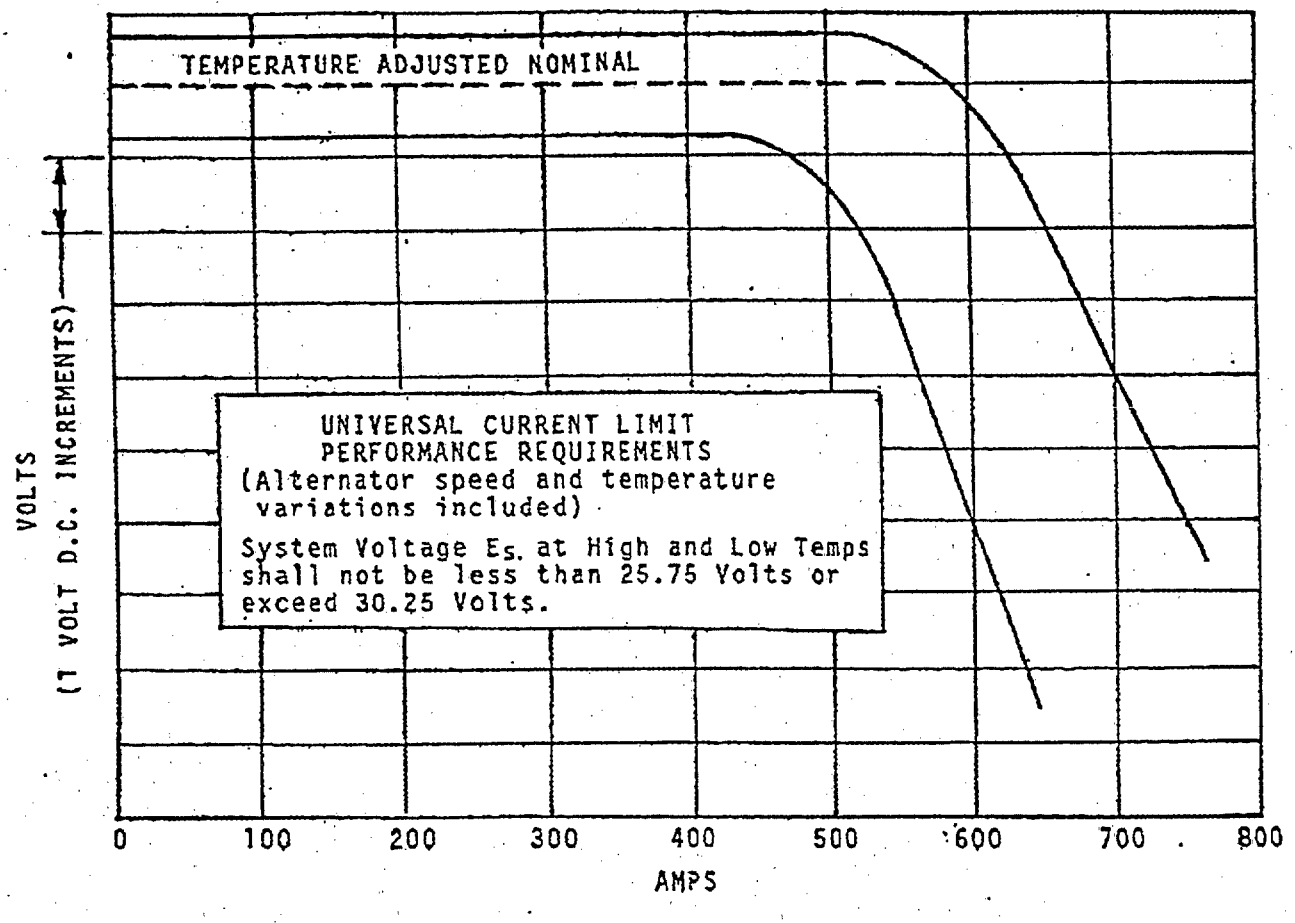


FIGURE 2. Universal current limit performance requirements.

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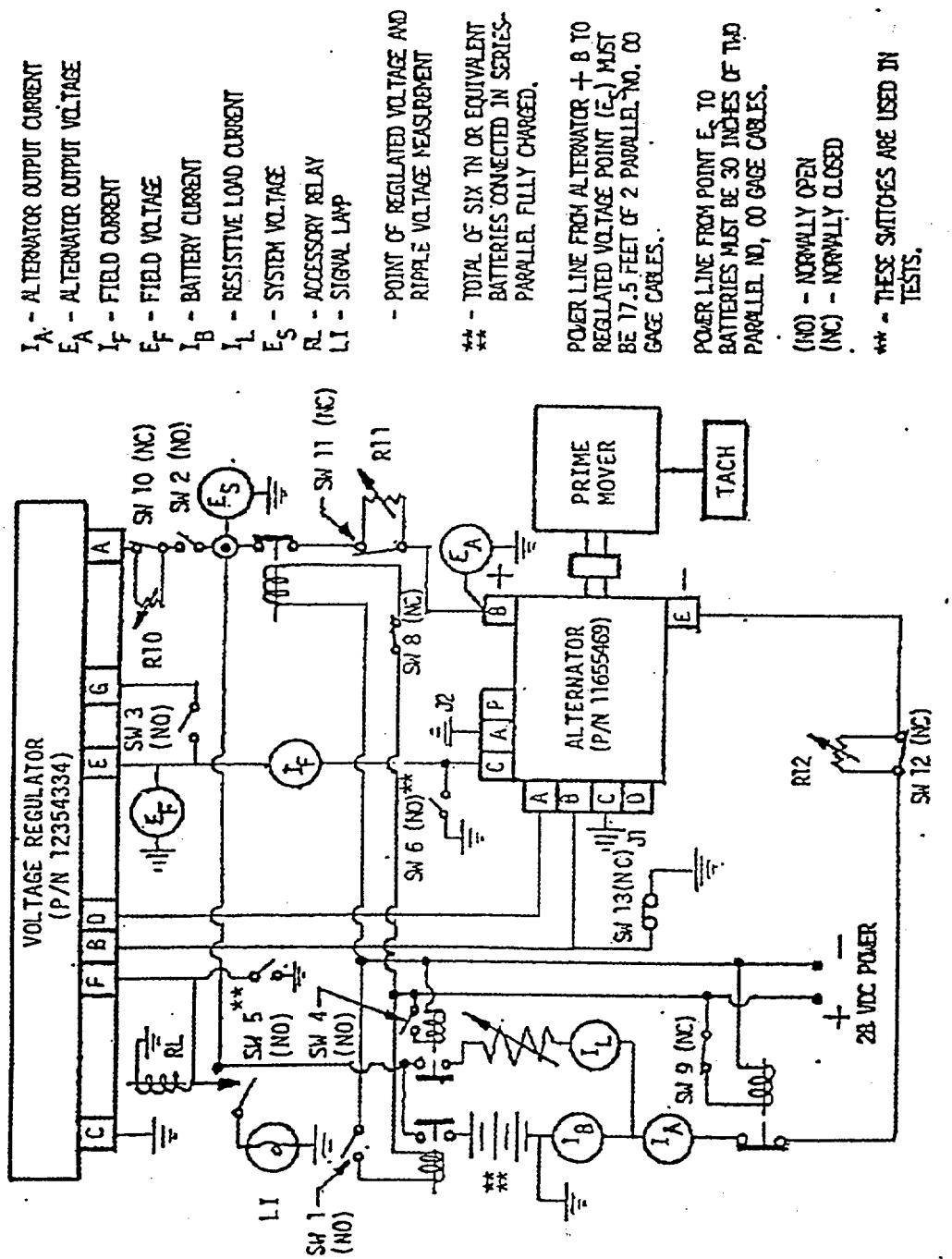


FIGURE 3. Voltage regulation test circuit.

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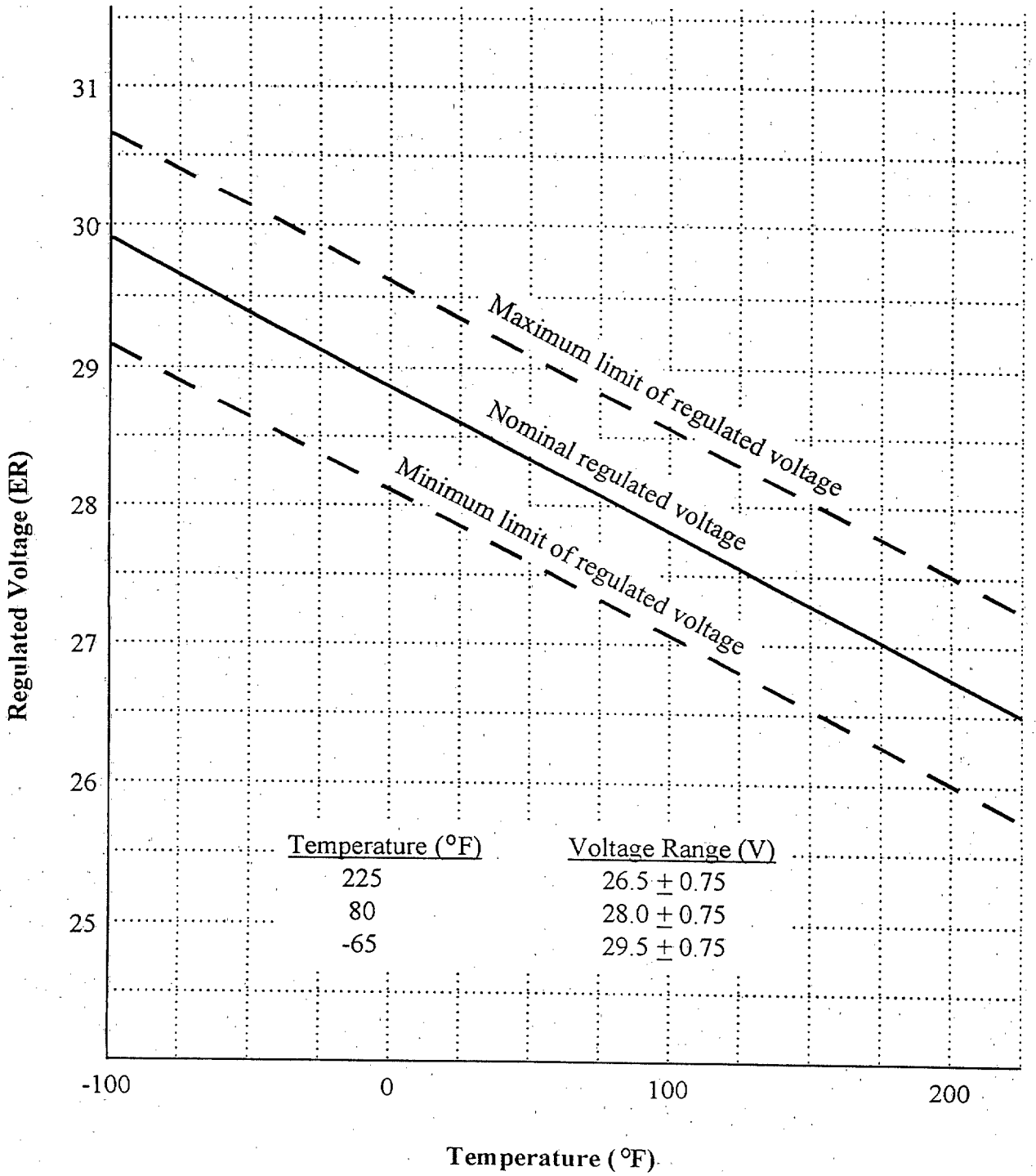


FIGURE 4. Voltage regulator temperature compensation (P/N 12354334).

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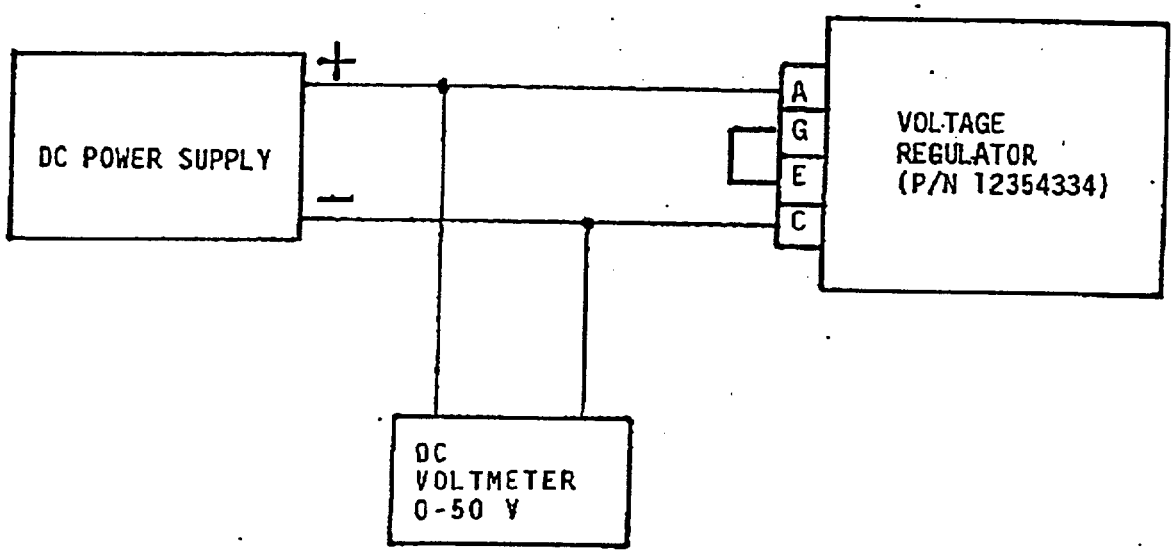


FIGURE 5. Overvoltage circuit test.

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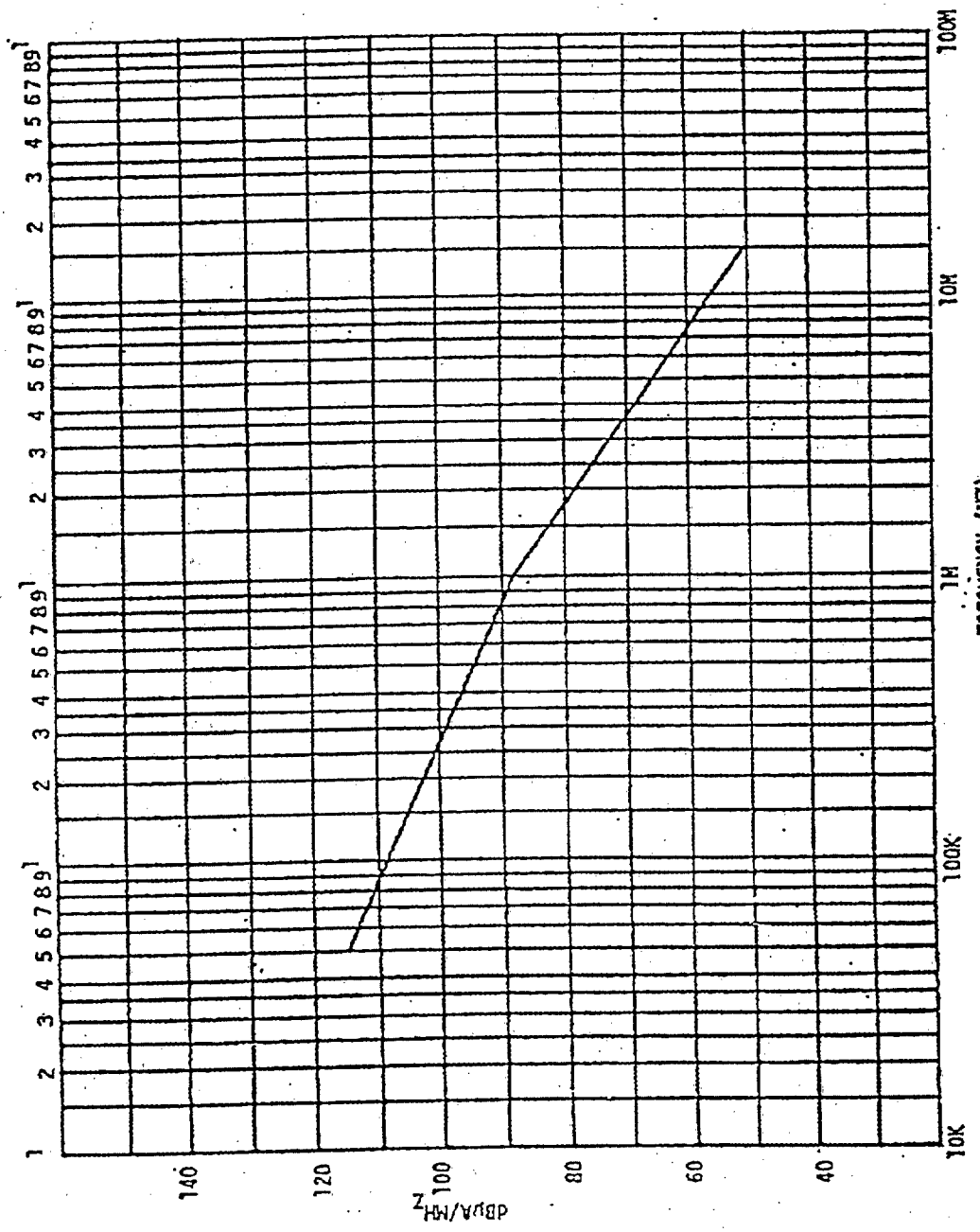


FIGURE 6. Broadband emission limits (conducted EMI).

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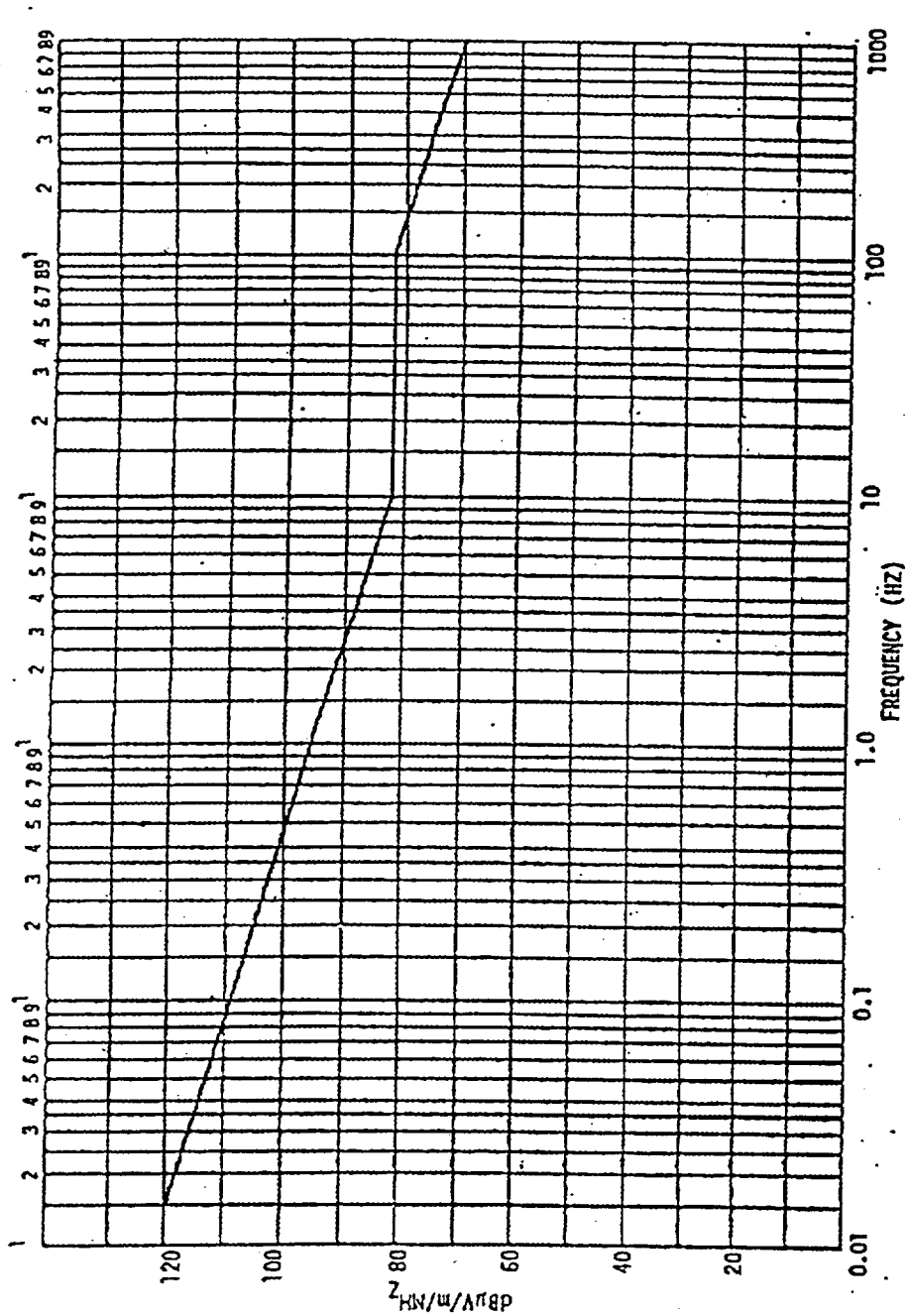
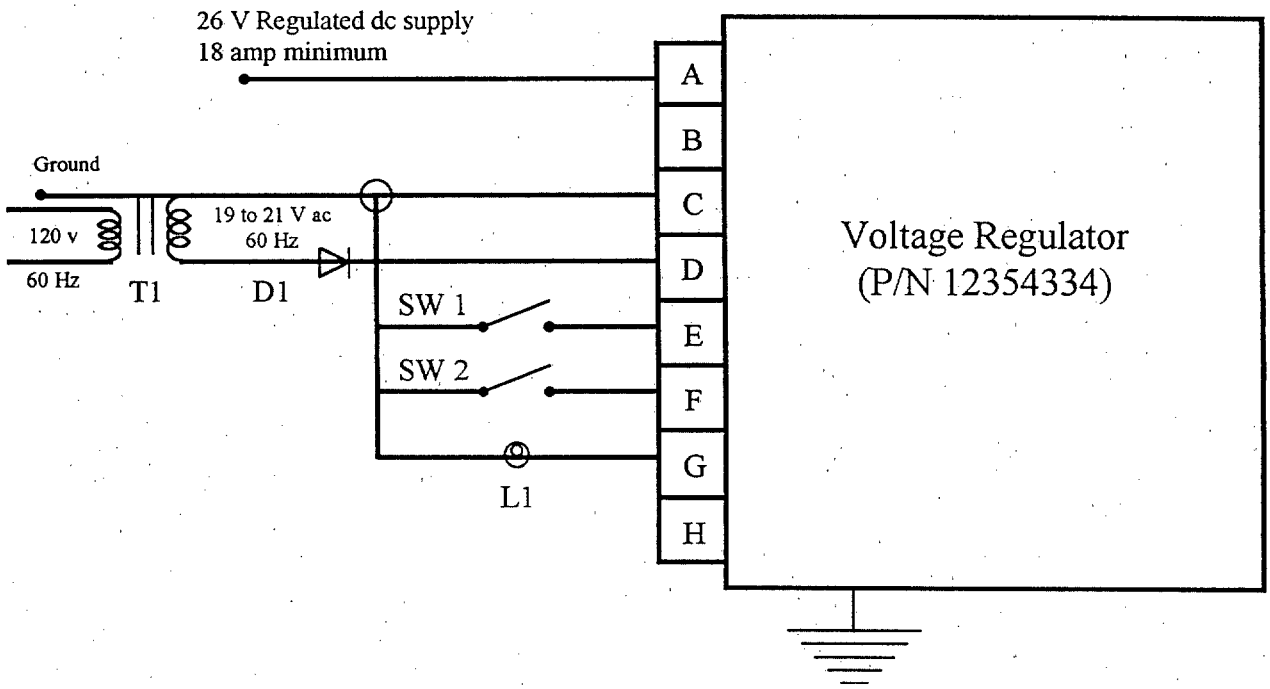


FIGURE 7. Broadband emission limits (radiated EMI).

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NOTES:

SW1 - Normally open single pole throw (S.P.S.T). Momentary SW18 amp minimum rating.
SW2 - Normally open S.P.S.T. momentary SW1 amp minimum rating.
L1 - 28 Vdc lamp - MS25237-327, or equivalent.
T1 - Transformer 60 Hz 19 to 21 V sec, 1 amp rating.
D1 - 3 amp 50 volt diode.

FIGURE 8. Short circuit protection.

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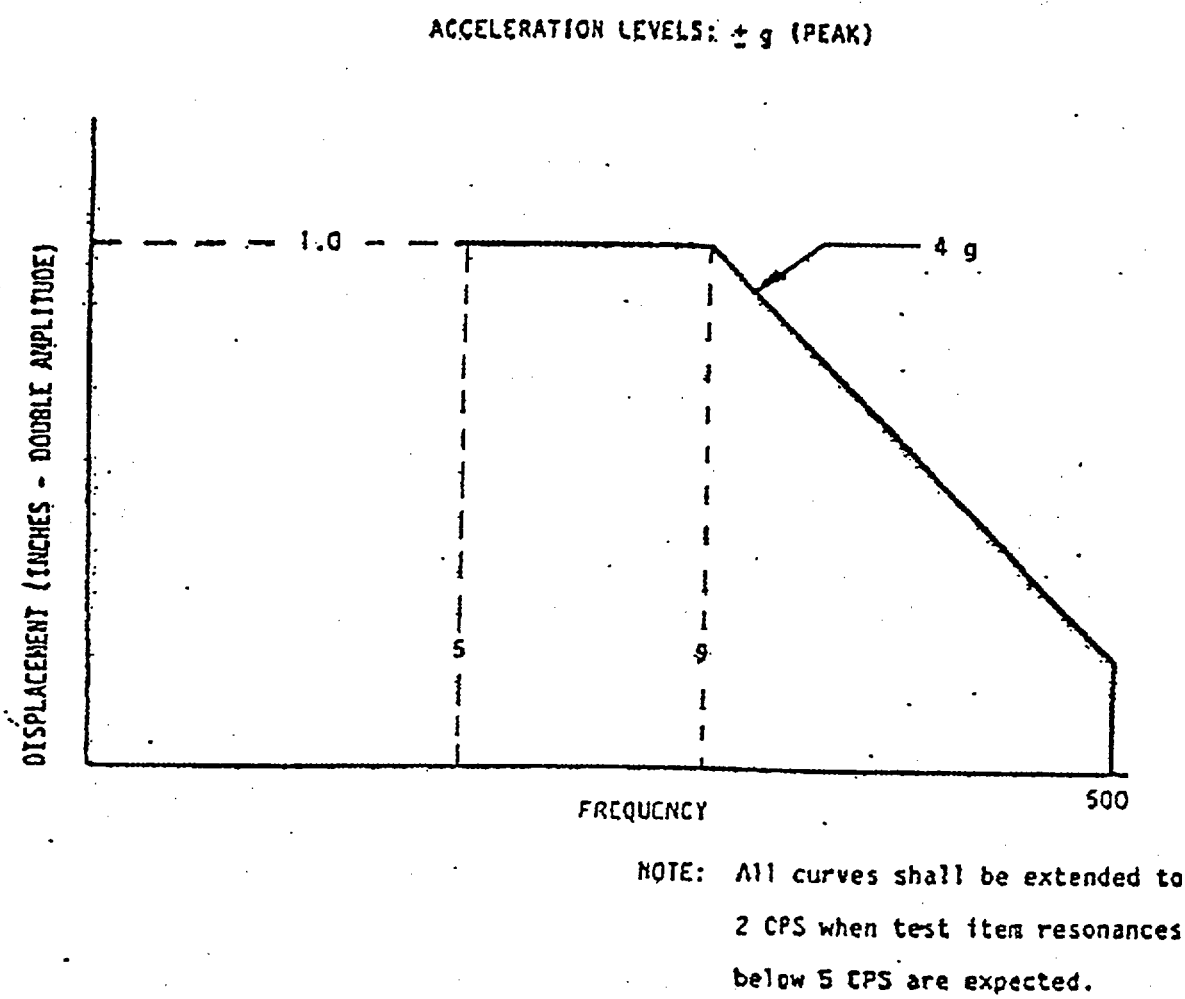


FIGURE 9. Vibration test curve for equipment installed in ground vehicles.

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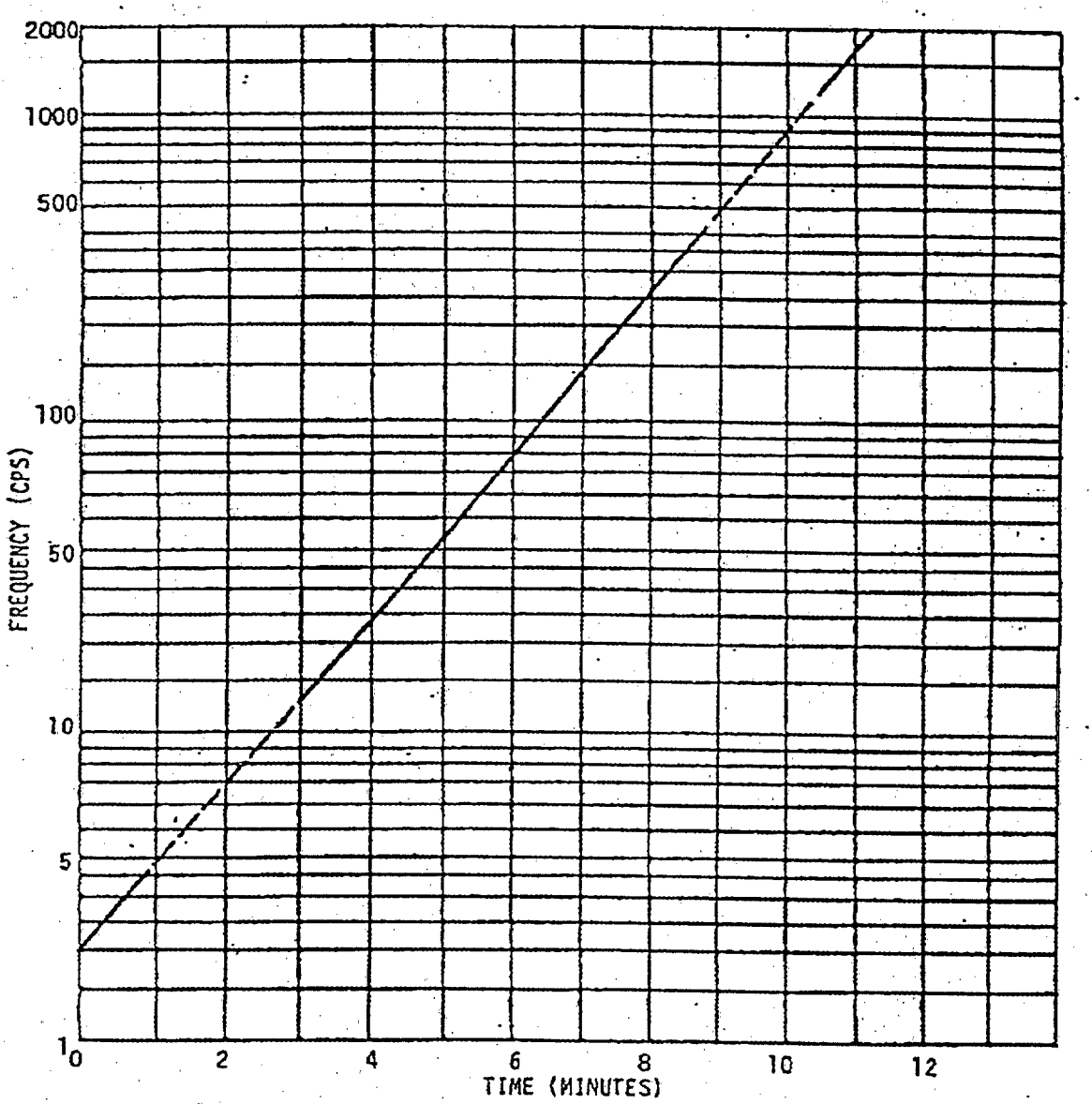
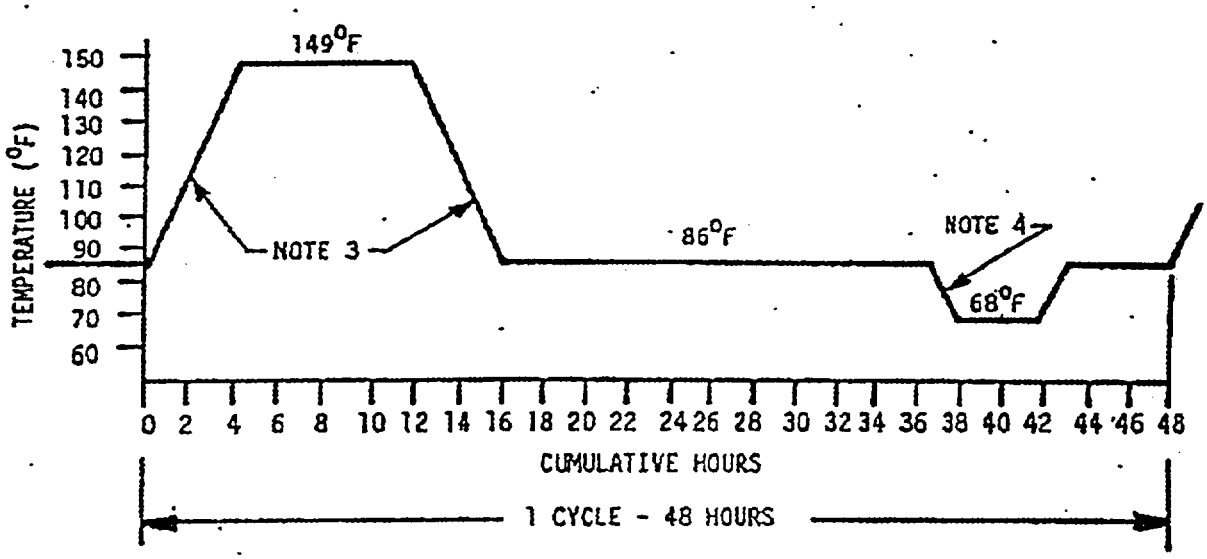


FIGURE 10. Logarithmic sweep.

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NOTES:

1. Tolerance during temperature change shall not be greater than 5°F.
2. Relative humidity shall be maintained at 94 ±4% at all times, except that during the descending temperature period, the relative humidity may be permitted to drop as low as 85%.
3. Rate of temperature change between 86° and 149°F shall not be less than 14.4°F per hour.
4. The temperature increase in this portion of the curve shall not be less than 18°F.

FIGURE 11. Humidity cycle.

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Preparing Activity:
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-62187A	2. DOCUMENT DATE (YYMMDD) 961230

3. DOCUMENT TITLE
REGULATOR, 650 AMP, SOLID STATE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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c. ADDRESS (Include Zip Code) Commander U.S. Army Tank-automotive and Armaments Command ATTN: AMSTA-TR-E/BLUE Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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